



Lab Report

Only for course Teacher						
		Needs Improvement	Developing	Sufficient	Above Average	Total Mark
Allocate mark & Percentage		25%	50%	75%	100%	
Understanding						
Analysis						
Implementation						
Report Writing						
Total obtained mark						
Comments						

Semester: Spring / Fall

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Batch: 46

Section: B

Course Code: CSE 124

Course Name: Object Oriented Programming

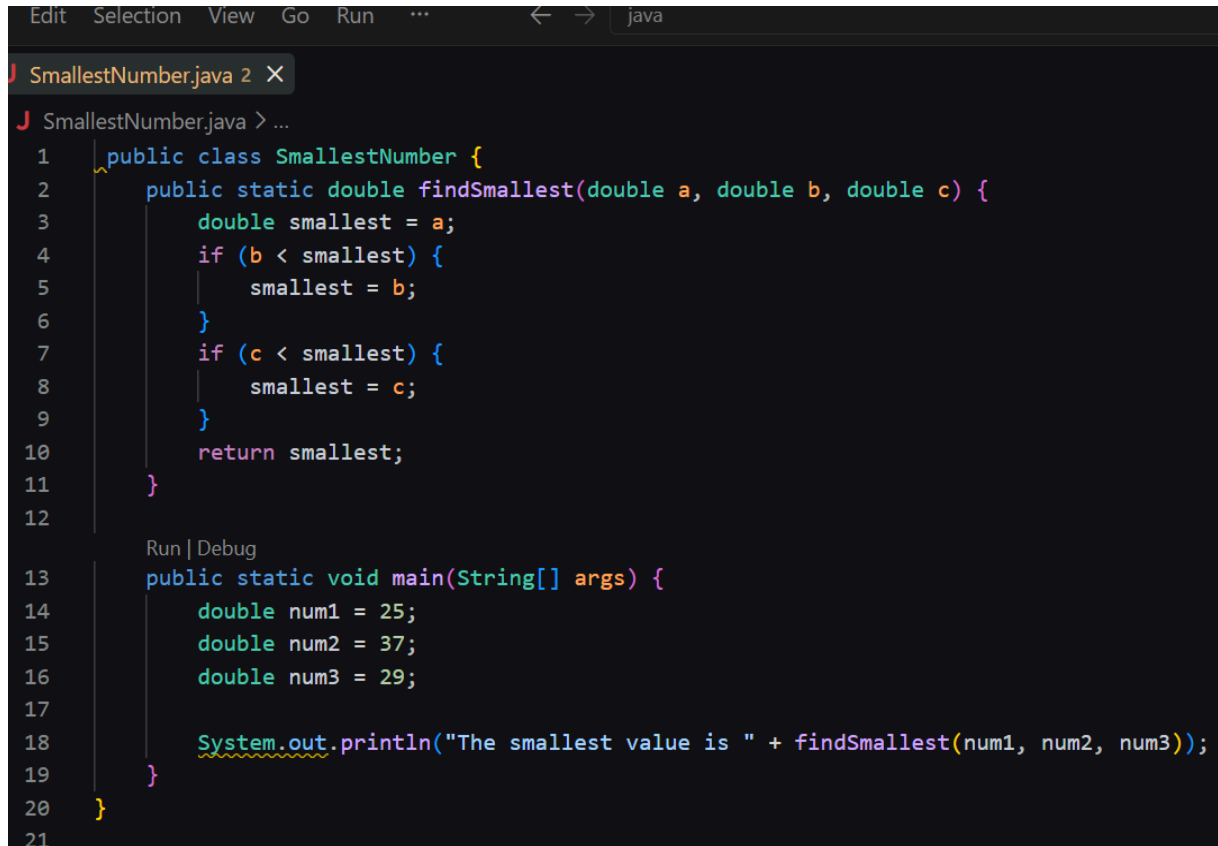
Course Teacher Name: Debabarata Mallick

Designation: Lecturer

Submission Date: 10 / 09 / 2026

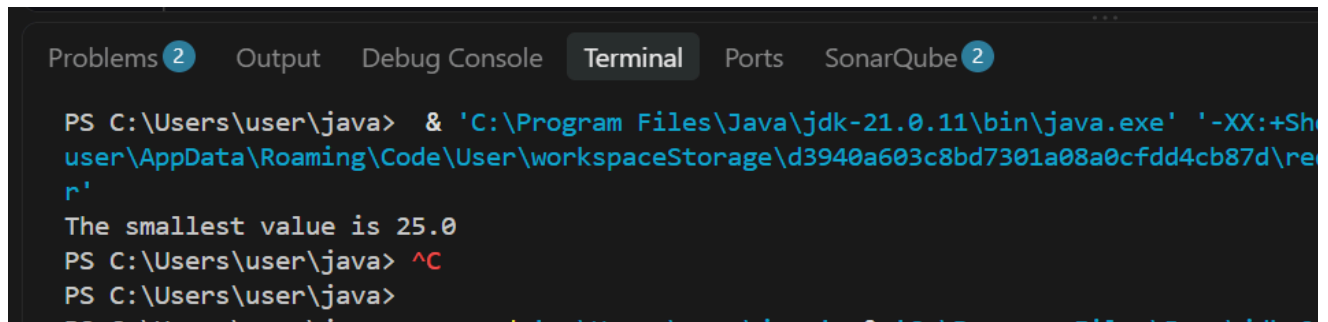
Experiment No : 1

Experiment Name : Find Smallest Number Among Three .



```
1 public class SmallestNumber {
2     public static double findSmallest(double a, double b, double c) {
3         double smallest = a;
4         if (b < smallest) {
5             smallest = b;
6         }
7         if (c < smallest) {
8             smallest = c;
9         }
10        return smallest;
11    }
12
13    public static void main(String[] args) {
14        double num1 = 25;
15        double num2 = 37;
16        double num3 = 29;
17
18        System.out.println("The smallest value is " + findSmallest(num1, num2, num3));
19    }
20 }
21
```

Output :



```
PS C:\Users\user\java> & 'C:\Program Files\Java\jdk-21.0.11\bin\java.exe' '-XX:+Sh
user\AppData\Roaming\Code\User\workspaceStorage\d3940a603c8bd7301a08a0cfdd4cb87d\re
r'
The smallest value is 25.0
PS C:\Users\user\java> ^C
PS C:\Users\user\java>
```

Discussion : I just used nested Math.min: Math.min(a, Math.min(b, c)). It simply compares the three numbers and returns the smallest one.

Experiment No : 2

Experiment Name : Compute Average of Three Numbers .

```
J AverageCalculator.java 2 X
J AverageCalculator.java > AverageCalculator
1 public class AverageCalculator {
2     public static double computeAverage(double a, double b, double c) {
3         return (a + b + c) / 3.0;
4     }
5
6     public static void main(String[] args) {
7         double num1 = 25;
8         double num2 = 45;
9         double num3 = 65;
10
11         System.out.println("The average value is " + computeAverage(num1, num2, num3));
12     }
13 }
```

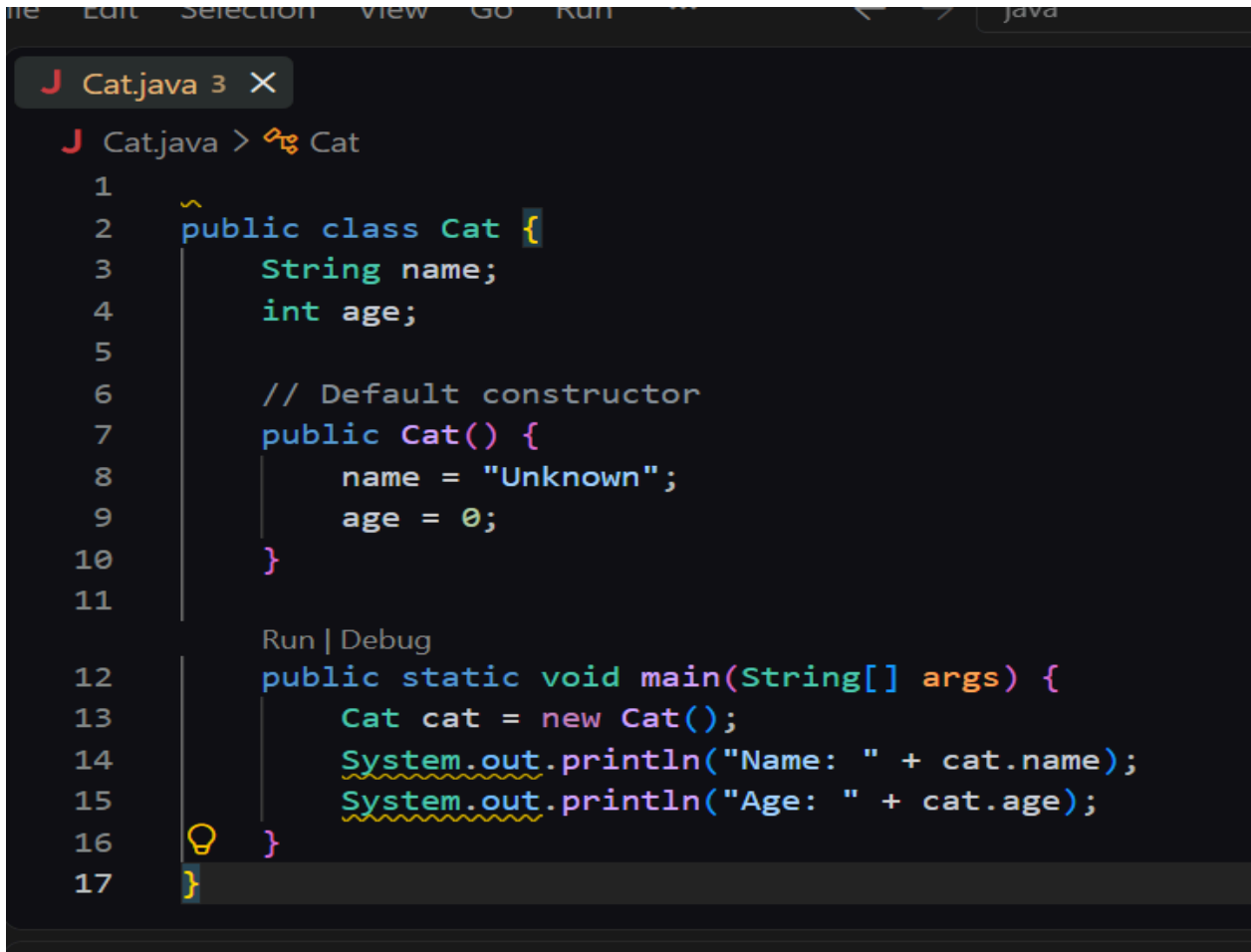
Output :

```
at AverageCalculator.main(SmallestNumber.java:6)
PS C:\Users\user\java> ^C
PS C:\Users\user\java>
PS C:\Users\user\java> c::; cd 'c:\Users\user\java'; & 'C:\Program Files\Java\jdk-21.0.
ionMessages' '-cp' 'C:\Users\user\AppData\Roaming\Code\User\workspaceStorage\d3940a603c
_c3870cc7\bin' 'AverageCalculator'
The average value is 45.0
PS C:\Users\user\java> █
```

Discussion : I calculated it by adding the three numbers and dividing by 3.0 so that the result stays in decimal form.

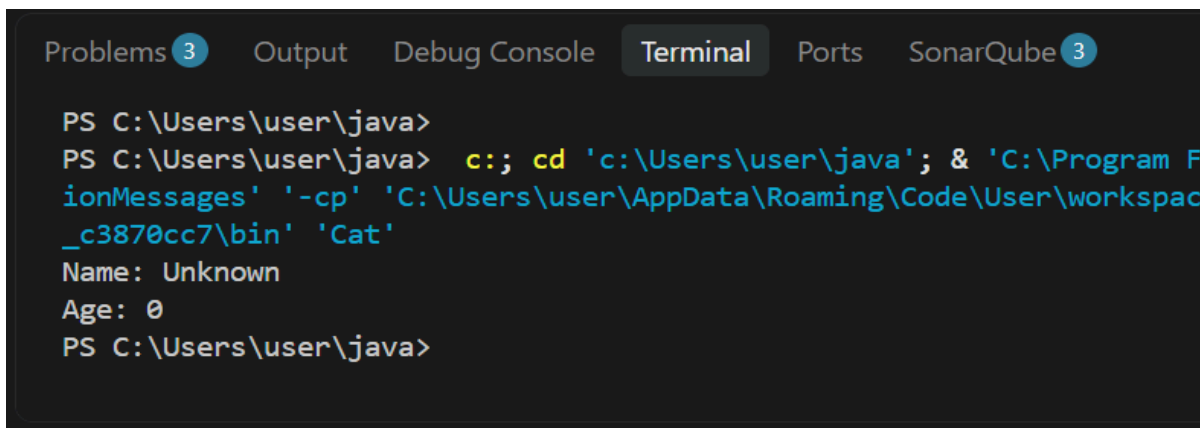
Experiment No : 3

Experiment Name : Default Constructor .



```
1  public class Cat {
2      String name;
3      int age;
4
5      // Default constructor
6      public Cat() {
7          name = "Unknown";
8          age = 0;
9      }
10
11      public static void main(String[] args) {
12          Cat cat = new Cat();
13          System.out.println("Name: " + cat.name);
14          System.out.println("Age: " + cat.age);
15      }
16  }
```

Output :

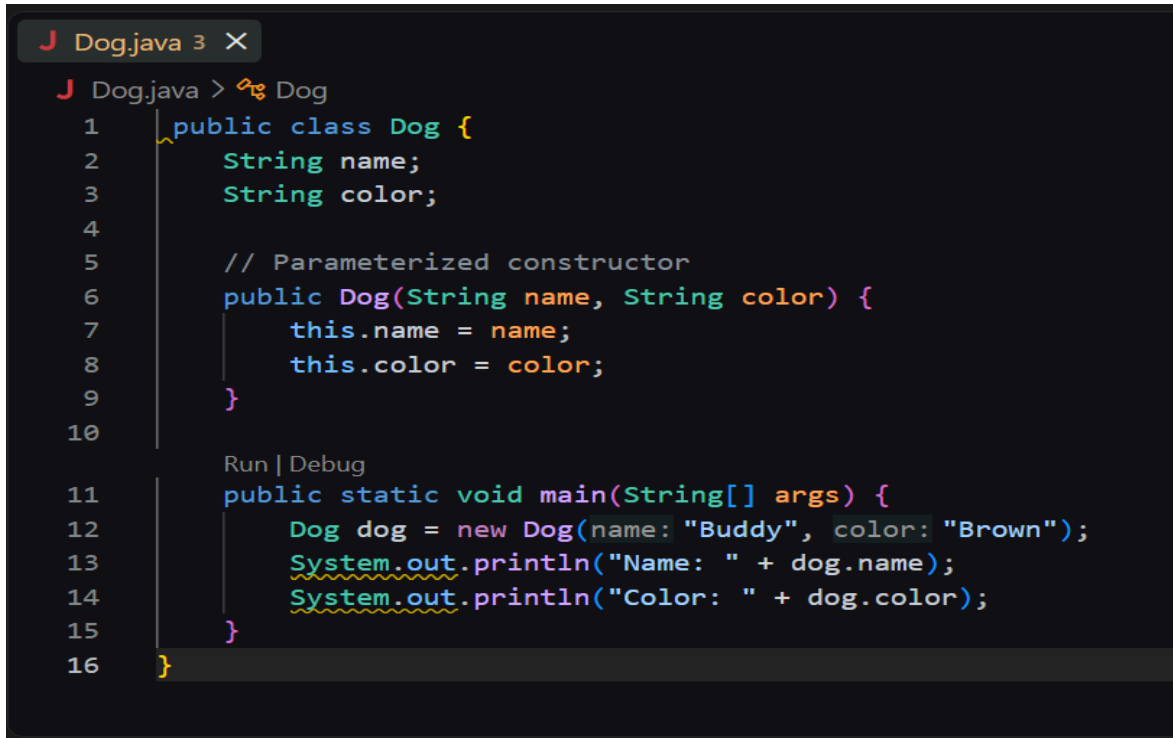


```
PS C:\Users\user\java>
PS C:\Users\user\java> c::; cd 'c:\Users\user\java'; & 'C:\Program Files\Java\jdk-11.0.10\bin\java.exe' -cp 'C:\Users\user\AppData\Roaming\Code\User\workspace _c3870cc7\bin' 'Cat'
Name: Unknown
Age: 0
PS C:\Users\user\java>
```

Discussion : I made a default constructor that sets name to "Unknown" and age to 0. When no values are passed, these default values are used.

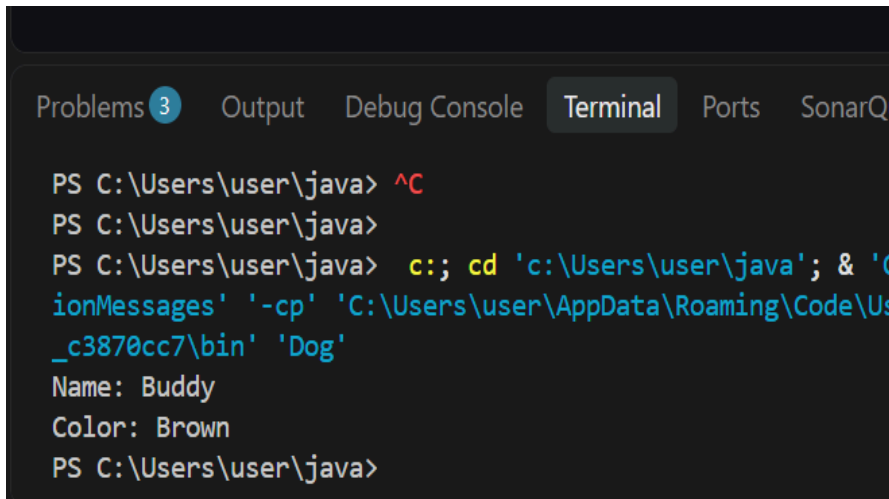
Experiment No : 4

Experiment Name : Parameterized Constructor .



```
J Dog.java 3 X
J Dog.java > Dog
1 public class Dog {
2     String name;
3     String color;
4
5     // Parameterized constructor
6     public Dog(String name, String color) {
7         this.name = name;
8         this.color = color;
9     }
10
11     Run | Debug
12     public static void main(String[] args) {
13         Dog dog = new Dog(name: "Buddy", color: "Brown");
14         System.out.println("Name: " + dog.name);
15         System.out.println("Color: " + dog.color);
16     }
}
```

Output :



```
Problems 3 Output Debug Console Terminal Ports SonarQ
PS C:\Users\user\java> ^C
PS C:\Users\user\java>
PS C:\Users\user\java> cd 'c:\Users\user\java'; & 'C:\Program Files\Java\jdk-11.0.10\bin\java.exe' -cp 'C:\Users\user\AppData\Roaming\Code\User\workspace_c3870cc7\bin' 'Dog'
Name: Buddy
Color: Brown
PS C:\Users\user\java>
```

Discussion : I created a parameterized constructor that takes name and color, then assigns them using `this.name` and `this.color`.

Experiment No : 5

Experiment Name : Text Formatter Utility .

```
J TextFormatter.java 6 X
J TextFormatter.java > TextFormatter
1 public class TextFormatter {
2
3     // Converts text to uppercase
4     public static String formatText(String text) {
5         return text.toUpperCase();
6     }
7
8     // Repeats the string the specified number of times, separated by space
9     public static String formatText(String text, int times) {
10        String result = "";
11        for (int i = 0; i < times; i++) {
12            result += text;
13            if (i < times - 1) {
14                result += " ";
15            }
16        }
17        return result;
18    }
19
20    // Attaches a prefix block to the front of the text
21    public static String formatText(String text, String prefix) {
22        return "[" + prefix + "] " + text;
23    }
24
25    Run | Debug
26    public static void main(String[] args) {
27        System.out.println(formatText(text: "hello"));
28        System.out.println(formatText(text: "Echo", times: 3));
29        System.out.println(formatText(text: "System updated", prefix: "INFO"));
30    }
31
32 0 12 Java: Ready SonarQube Watch
```

```
29 }
30 }

Problems 12 Output Debug Console Terminal Ports SonarQube 12

PS C:\Users\user\java> c:; cd 'c:\Users\user\java'; & 'C:\Program
ionMessages' '-cp' 'C:\Users\user\AppData\Roaming\Code\User\worksp
_c3870cc7\bin' 'TextFormatter'
HELLO
Echo Echo Echo
[INFO] System updated
PS C:\Users\user\java>
```

Discussion : I used method overloading.

- One method converts the text to uppercase.
- Second one repeats the text the given number of times with spaces.
- Third one adds a prefix like [INFO] in front of the text.

Experiment No : 6

Experiment Name : Notification Dispatcher .

```
J NotificationSystem.java 4 X
J NotificationSystem.java > NotificationSystem
1 public class NotificationSystem {
2
3     // Method 1: Default system broadcast
4     public void sendNotification(String message) {
5         System.out.println("Broadcast Alert: " + message);
6     }
7
8     // Method 2: Targeted email alert
9     public void sendNotification(String message, String email) {
10        System.out.println("Email sent to " + email + " -> " + message);
11    }
12
13    // Method 3: SMS text alert
14    public void sendNotification(String message, long phoneNumber) {
15        System.out.println("SMS sent to +" + phoneNumber + " -> " + message);
16    }
17
18    Run | Debug
19    public static void main(String[] args) {
20        NotificationSystem ns = new NotificationSystem();
21
22        ns.sendNotification(message: "Server restart in 5 mins.");
23        ns.sendNotification(message: "Your OTP is 4432", email: "user@test.com");
24        ns.sendNotification(message: "Package delivered!", phoneNumber: 15550199L);
25    }
```

Output :

```
PS C:\Users\user\java> ^C
PS C:\Users\user\java> c::; cd 'c:\Users\user\java'; & 'C:\Program Files\Java\j
ionMessages' '-cp' 'C:\Users\user\AppData\Roaming\Code\User\workspaceStorage\d3
_c3870cc7\bin' 'NotificationSystem'
Broadcast Alert: Server restart in 5 mins.
Email sent to user@test.com -> Your OTP is 4432
SMS sent to +15550199 -> Package delivered!
PS C:\Users\user\java>
```

Discussion : Again used method overloading with the same method name `sendNotification`:

- **One parameter** → broadcasts the message
- **Two parameters (message + email)** → sends email
- **Two parameters (message + phone number)** → sends SMS